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United States Patent	12416228
Kind Code	B2
Date of Patent	September 16, 2025
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Artificial intelligence based hydraulic fracturing system monitoring and control

Abstract

A system monitors operation of a component in a hydraulic fracturing fleet. A sensor exposed to an external environment of the component is configured to detect external indicia of the operation of the component. Memory stores an artificial intelligence (AI) model, the AI model being trained to monitor the operation of the component in the system. One or more processors are operatively coupled to the memory and the sensor. The one or more processors are configured to obtain data of the external indicia detected with the sensor; input the obtained data into the AI model; detect, with the AI model and based on the input data of the external indicia, one of a plurality of predetermined states corresponding to the operation of the component; and perform a predetermined function based on the detected one of the plurality of predetermined states.

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Appl. No.: 18/425851

Filed: January 29, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240318533 A1	Sep. 26, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: **E21B43/26** (20060101); **F04B49/06** (20060101); **G06N5/04** (20230101); **G06N20/00** (20190101)

U.S. Cl.:

CPC **E21B43/2607** (20200501); **F04B49/065** (20130101); **G06N5/04** (20130101); **G06N20/00** (20190101); **E21B2200/22** (20200501); **F04B2201/0208** (20130101)

Field of Classification Search

CPC: E21B (2200/22); E21B (43/2607); E21B (4/00); E21B (27/00); E21B (11/00); F04B (15/02); F04B (2201/0208); F04B (23/06); F04B (47/02); F04B (49/007); F04B (49/065); F04B (3/00); F04B (2201/00); F04B (2205/00); G06N (20/00); G06N (5/04); G06N (3/004); G06N (3/02); G06N (3/12); G06N (3/00)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/938,759, filed Jul. 24, 2020, which claims the benefit of U.S. Provisional Application No. 62/879,239 filed Jul. 26, 2019, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates generally to hydraulic fracturing. More particularly, but not by way of limitation, this disclosure relates to utilizing artificial intelligence (AI) techniques in hydraulic fracturing operations to monitor and control hydraulic fracturing equipment on one or more transports for different use cases.

BACKGROUND

(2) Hydraulic fracturing has been commonly used by the oil and gas industry to stimulate production of hydrocarbon wells, such as oil and/or gas wells. Hydraulic fracturing, sometimes called “fracing” or “fracking,” is the process of injecting fracturing fluid, which is typically a mixture of water, sand, and chemicals, into the subsurface to fracture the subsurface geological formations and release otherwise encapsulated hydrocarbon reserves. The fracturing fluid is typically pumped into a wellbore at a relatively high pressure sufficient to cause fissures within the underground geological formations. Specifically, once inside the wellbore, the pressurized fracturing fluid is pressure pumped down and then out into the subsurface geological formation to fracture the underground formation. A fluid mixture that may include water, various chemical additives, and proppants (e.g., sand or ceramic materials) can be pumped into the underground formation to fracture and promote the extraction of the hydrocarbon reserves, such as oil and/or gas. For example, the fracturing fluid may comprise a liquid petroleum gas, linear gelled water, gelled water, gelled oil, slick water, slick oil, poly emulsion, foam/emulsion, liquid carbon dioxide, nitrogen gas, and/or binary fluid and acid.

(3) Implementing large-scale fracturing operations at well sites typically require extensive investment in equipment, labor, and fuel. For instance, a typical fracturing operation uses a variety of fracturing equipment, numerous personnel to operate and maintain the fracturing equipment, large amounts of fuel to power the fracturing operations, and large volumes of fracturing fluids. As such, planning for fracturing operations is often complex and encompasses a variety of logistical challenges that include minimizing the on-site area or “footprint” of the fracturing operations, providing adequate power and/or fuel to continuously power the fracturing operations, increasing the efficiency of the hydraulic fracturing equipment, and reducing any environmental impact resulting from fracturing operations. Thus, numerous innovations and improvements of existing fracturing technology are needed to address the variety of complex and logistical challenges faced in today's fracturing operations.

SUMMARY

(4) The following presents a simplified summary of the disclosed subject matter in order to provide a basic understanding of some aspects of the subject matter disclosed herein. This summary is not an exhaustive overview of the technology disclosed herein. It is not intended to identify key or critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is discussed later.

(5) In one embodiment, a method includes obtaining data of at least one external indicia of operation of at least one component in a hydraulic fracturing fleet, wherein the data is detected with

at least one sensor; inputting the obtained data into an Artificial Intelligence (AI) model, the AI model being trained to monitor the operation of the at least one component in the hydraulic fracturing fleet; detecting, with the AI model and based on the input data of the at least one external indicia, one of a plurality of predetermined states corresponding to the operation of the at least one component; and performing a predetermined function based on the detected one of the plurality of predetermined states.

(6) In another embodiment, the method may be embodied in computer executable program code and stored in a non-transitory storage device. In yet another embodiment, the method may be implemented on a system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 is a schematic diagram of a mobile hydraulic fracturing system operating at a well site, in accordance with one or more embodiments.

(3) FIG. 2A is a schematic diagram of an embodiment of a fracturing pump transport that is able to engage or disengage one or more pumps from a prime mover.

(4) FIG. 2B is a schematic diagram of an embodiment of the fracturing pump transport that is able to engage or disengage one or more pumps from the prime mover.

(5) FIG. 3A illustrates a cross-section view of an exemplary pump provided on the fracturing pump transport in accordance with one or more embodiments.

(6) FIG. 3B illustrates a cross-section view of a fluid end assembly of the exemplary pump provided on the fracturing pump transport in accordance with one or more embodiments.

(7) FIG. 4 illustrates an example of an image of the fluid end assembly that is captured by a sensor in accordance with one or more embodiments.

(8) FIG. 5 illustrates another example of an image of the fluid end assembly that is captured by a sensor in accordance with one or more embodiments.

(9) FIG. 6 is a block diagram of an embodiment of an AI system that monitors, controls, and communicates with one or more control systems associated with one or more transports of the hydraulic fracturing system.

(10) FIG. 7 is a dataflow graph illustrating an exemplary technique for creating an AI model that recognizes patterns and learns over time, in accordance with one or more embodiments.

(11) FIG. 8A is a schematic diagram of an embodiment of a blender transport that includes an electric blender.

(12) FIG. 8B is a schematic diagram of an embodiment of a blender transport that includes an electric blender.

(13) FIG. 9A is a schematic diagram of an embodiment of a blender transport that includes an electric blender with enclosed mixer hoppers.

(14) FIG. 9B is a schematic diagram of an embodiment of a blender transport that includes an electric blender with enclosed mixer hoppers.

(15) FIG. 10 is a flowchart illustrating a technique for creating and implementing the AI model in accordance with one or more embodiments.

(16) FIG. 11 shows an illustrative simplified block diagram of a computing system that includes the AI model in accordance with one or more embodiments.

(17) FIG. 12 shows an illustrative block diagram of a computer network in accordance with one or more embodiments.

(18) While certain embodiments will be described in connection with the illustrative embodiments shown herein, the invention is not limited to those embodiments. On the contrary, all alternatives, modifications, and equivalents are included within the spirit and scope of the invention as defined by the claims. In the drawings, which are not to scale, the same reference numerals are used throughout the description and in the drawing figures for components and elements having the same structure, and primed reference numerals are used for components and elements having a similar function and construction to those components and elements having the same unprimed reference numerals.

DESCRIPTION

(19) In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the inventive concept. In the interest of clarity, not all features of an actual implementation are described. Moreover, the language used in this disclosure has been principally selected for readability and instructional purposes, and may not have been selected to delineate or circumscribe the inventive subject matter, resort to the claims being necessary to determine such inventive subject matter. Reference in this disclosure to “one embodiment” or to “an embodiment” or “another embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention, and multiple references to “one embodiment” or “an embodiment” or “another embodiment” should not be understood as necessarily all referring to the same embodiment.

(20) The terms “a,” “an,” and “the” are not intended to refer to a singular entity unless explicitly so defined, but include the general class of which a specific example may be used for illustration. The use of the terms “a” or “an” may therefore mean any number that is at least one, including “one,” “one or more,” “at least one,” and “one or more than one.” The term “or” means any of the alternatives and any combination of the alternatives, including all of the alternatives, unless the alternatives are explicitly indicated as mutually exclusive. The phrase “at least one of” when combined with a list of items, means a single item from the list or any combination of items in the list. The phrase does not require all of the listed items unless explicitly so defined.

(21) As used herein, the term “transport” refers to any transportation assembly, including, but not limited to, a trailer, truck, skid, and/or barge used to transport relatively heavy structures, such as a mobile gas turbine generator.

(22) As used herein, the term “trailer” refers to a transportation assembly used to transport relatively heavy structures, such as a mobile gas turbine generator that can be attached and/or detached from a transportation vehicle used to pull or move the trailer. In one embodiment, the trailer may include the mounts and manifold systems to connect the trailer to other equipment.

(23) As used herein, the term “lay-down trailer” refers to a trailer that includes two sections with different vertical heights. One of the sections or the upper section is positioned at or above the trailer axles and another section or the lower section is positioned at or below the trailer axles. In one embodiment the main trailer beams of the lay-down trailer may be resting on the ground when in operational mode and/or when uncoupled from a transportation vehicle, such as a tractor.

(24) As used herein, the term “gas turbine generator” refers to both the gas turbine and the generator sections of a gas-turbine generator transport (e.g., power generation transport, mobile source of electricity, and the like). The gas turbine generator receives hydrocarbon fuel, such as natural gas, and converts the hydrocarbon fuel into electricity.

(25) As used herein, the term “inlet plenum” may be interchanged and generally referred to as “inlet,” “air intake,” and “intake plenum,” throughout this disclosure. Additionally, the term “exhaust collector” may be interchanged throughout and generally referred to as “exhaust diffuser” and “exhaust plenum” throughout this disclosure.

(26) As used herein, the term “gas turbine inlet filter” may be interchanged and generally referred to as “inlet filter” and “inlet filter assembly.” The term “air inlet filter housing” may also be

interchanged and generally referred to as “filter housing” and “air filter assembly housing” throughout this disclosure. Furthermore, the term “exhaust stack” may also be interchanged and generally referred to as “turbine exhaust stack” throughout this disclosure.

(27) This disclosure pertains to an AI system that monitors, controls, and communicates with one or more control systems controlling hydraulic fracturing equipment (e.g., apparatuses, components, and the like) on one or more transports of a hydraulic fracturing fleet. Techniques disclosed herein utilize artificial intelligence techniques to detect predetermined states (e.g., different fluid end packing failure states, blender or hopper states, states related to high-pressure leak on iron between pumps and well head, and the like) that may impact operation of hydraulic fracturing equipment (e.g., apparatus, component, and the like) on one or more transports of a hydraulic fracturing fleet or system. The AI system may then perform predetermined operations or functions (e.g., shut down affected hydraulic fracturing apparatus, notify operator, and the like) based on the detected state.

(28) In one embodiment, one or more AI models (e.g., machine learning model, deep learning model, and the like) may be trained to detect predetermined states based on corresponding sensor data (e.g., image data, thermal imaging data, audio data, other sensor data, combination of various sensor data, and the like) of one or more apparatuses (e.g., equipment, components, and the like) on one or more hydraulic fracturing transports of the hydraulic fracturing system. The AI models may correspond to respective use cases for AI-based state detection on the hydraulic fracturing system. For example, an AI model may be trained based on sensor data (e.g., one or more of image data, thermal imaging data, audio data, other sensor data, and the like) associated with a fluid end assembly use case. In the fluid end assembly use case, sensor data of a fluid end assembly of a pump on a fracturing pump transport may be used to detect predetermined states, e.g., a state when the pump starts overheating, a state when it starts smoking, and a state when it begins to leak due to high-pressure sand laden fluid for hydraulic fracturing cutting a groove into a packing bore of the fluid end assembly. The trained AI model for the fluid end use case may then be deployed on the AI system to automatically detect the predetermined states for the fluid end packing use case for each pump of each frac pump transport of the fleet based on corresponding sensor data captured by one or more sensors in real-time, and perform predetermined operations or functions (e.g., shutdown pump, notify operator, alert user in “danger zone” and the like) for the respective fluid end assembly of the pump based on the determined state thereof.

(29) In the fluid end assembly use case, during hydraulic fracturing operations, in the event of a packing failure at the fluid end, if the pump operator does not shut down the pump in a very rapid manner (e.g., within order of seconds), the high-pressure sand laden hydraulic fracturing fluid can cut a groove into the packing bore of the fluid end of the pump, and make the fluid end un-useable without costly and sometimes unreliable repair. This also leads to downtime and sacrifices job performance. By implementing for this use case, an AI model that is trained to automatically and dynamically detect fluid end packing failure states (e.g., state in which fluid end packing begins to smoke, state in which fluid end packing begins to overheat, state in which fluid end starts to leak high-pressure fluid, a normal operating state, and the like), and perform predetermined operations or functions (e.g., shutdown pump, notify operator, change lubrication rate of the pump, alert user in “danger zone”, and the like) based on the detected state, significant damage to the pump can be prevented and downtime can be reduced.

(30) Additional AI models may similarly be trained for other use cases on the hydraulic fracturing system. That is, other AI models may be trained based on sensor data associated with other hydraulic fracturing equipment (e.g., apparatuses, components, and the like) on one or more transports of the hydraulic fracturing system to detect corresponding predetermined states (e.g., states related to a blender or a hopper on a blender transport, states related to high-pressure leak on iron between pump and well head) for other use cases. The additional AI models may similarly be deployed on the AI system for the additional use cases of the AI system to automatically detect predetermined states the AI models are respectively trained to detect based on sensor data captured

by the one or more sensors in real-time, and perform predetermined operations or functions (e.g., shutdown affected components or apparatus, notify operator, alert user in “danger zone”, and the like) based on the detected states. Examples of other use cases for the AI system implemented on the hydraulic fracturing system may include use cases associated with any component of the fracturing fleet that includes reciprocating or rotating equipment (e.g., chemical additive unit, indoor mix tubs on blender transport, hopper on blender transport, batch tank on hydration unit, and the like) where it may be beneficial to have sensors (e.g., cameras, thermal imaging system, and the like) installed for monitoring and control.

(31) The AI system disclosed herein may be configured to monitor and control different fracturing components located at well sites. The different fracturing apparatuses, which include, but are not limited to, a blender, hydration unit, sand handling equipment, chemical additive system, fracturing pumps, prime mover, and mobile source of electricity (e.g., power generation transport), may be configured to operate remotely via a control network system (e.g., system including an AI system) that monitors and controls the fracturing equipment using a network topology, such as an Ethernet ring topology or star topology network. The control network system may remove the need for implementing multiple control systems located on and/or in close proximity to the fracturing transports. Instead, a designated location, such as a data van and/or a remote location away from the vicinity of the fracturing equipment may remotely control the hydraulic fracturing equipment on the one or more transports.

(32) FIG. 1 is a schematic diagram an embodiment of well site **100** that comprises wellhead **101** and mobile fracturing system **103** (e.g., hydraulic fracturing fleet or system). Generally, mobile fracturing system **103** may perform fracturing operations to complete a well and/or transform a drilled well into a production well. For example, well site **100** may be a site where operators are in the process of drilling and completing a well. Operators may start the well completion process with drilling, running production casing, and cementing within the wellbore. The operators may also insert a variety of downhole tools into the wellbore and/or as part of a tool string used to drill the wellbore. After the operators drill the well to a certain depth, a horizontal portion of the well may also be drilled and subsequently encased in cement. The operators may subsequently remove the rig, and mobile fracturing system **103** may be moved onto well site **100** to perform fracturing operations that force relatively high-pressure fracturing fluid through wellhead **101** into subsurface geological formations to create fissures and cracks within the rock. Fracturing system **103** may be moved off well site **100** once the operators complete fracturing operations. Typically, fracturing operations for well site **100** may last several days.

(33) To provide an environmentally cleaner and more transportable fracturing fleet, mobile fracturing system **103** may comprise power generation transport **102** (e.g., mobile source of electricity) configured to generate electricity by converting hydrocarbon fuel, such as natural gas, obtained from one or more other sources (e.g., a producing wellhead) at well site **100**, from a remote offsite location, and/or another relatively convenient location near power generation transport **102**. Improving mobility of mobile fracturing system **103** may be beneficial because fracturing operations at a well site typically last for several days and the fracturing equipment is subsequently removed from the well site after completing fracturing operation. Rather than using fuel that significantly impacts air quality (e.g., diesel fuel) as a source of power and/or receiving electric power from a grid or other type of stationary power generation facility (e.g., located at the well site or offsite), mobile fracturing system **103** utilizes power generation transport **102** as a power source that burns cleaner while being transportable along with other fracturing equipment. The generated electricity from power generation transport **102** may be supplied to fracturing equipment to power fracturing operations at one or more well sites, or to other equipment in various types of applications requiring mobile electric power generation.

(34) As shown in FIG. 1, power generation transport **102** may be implemented as a single power generation transport in order to reduce the well site footprint and provide the ability for operators to

easily move power generation transport **102** to different well sites and/or different fracturing jobs and/or different physical locations. In other embodiments, power generation transport **102** may be implemented using two or more transports. For example, power generation transport **102** may be implemented using a two-transport design in which a first transport may be a power generation transport comprising a turbine (e.g., gas turbine) and a generator, and a second transport may be an inlet and exhaust transport that comprises an inlet plenum providing filtered combustion air for the turbine, and an exhaust stack that securely provides an exhaust system for exhaust air from the turbine.

(35) In addition to power generation transport **102**, mobile fracturing system **103** may include switch gear transport **112**, at least one blender transport **110**, at least one data van **114**, and one or more fracturing pump transports **108** that deliver fracturing fluid through wellhead **101** to subsurface geological formations. Switch gear transport **112** may receive electricity generated by power generation transport **102** via one or more electrical connections. In one embodiment, switch gear transport **112** may use 13.8 kilovolts (kV) electrical connections to receive power from power generation transport **102**. Switch gear transport **112** may comprise a plurality of electrical disconnect switches, fuses, transformers, and/or circuit protectors to protect the fracturing equipment. The switch gear transport **112** may transfer the electricity received from power generation transport **102** to the electrically connected fracturing equipment of mobile fracturing system **103**. Switch gear transport **112** may further comprise a control system to control, monitor, and provide power to the electrically connected fracturing equipment.

(36) In one embodiment, switch gear transport **112** may receive a 13.8 kV electrical connection and step down the voltage to 4.8 kV, which is provided to other fracturing equipment, such as fracturing pump transport **108**, blender transport **110**, sand storage and conveyor, hydration equipment, chemical equipment, data van **114**, lighting equipment, sensor equipment (e.g., image sensor, thermal imaging sensor, light sensor, sound sensor, and the like) and any additional auxiliary equipment used for the fracturing operations. Switch gear transport **112** may step down the voltage to 4.8 kV rather than other voltage levels, such as 600 V, in order to reduce cable size for the electrical connections and the amount of cabling used to connect mobile fracturing system **103**. In another embodiment, the voltage step down operation may be performed further downstream from switch gear transport **112**. For example, switch gear transport **112** may provide the received 13.8 kV electrical connection directly to the fracturing pump transport **108**. The voltage step down operation may then be performed on fracturing pump transport **108**. The control system of switch gear transport **112** may be configured to connect to the control network system (e.g., AI system) such that switch gear transport **112** may be monitored and/or controlled from a distant location, such as data van **114** or some other type of control center.

(37) Fracturing pump transport **108** may receive the electric power from switch gear transport **112** to power a prime mover. The prime mover converts electric power to mechanical power for driving one or more pumps. In one embodiment, the prime mover may be a dual shaft electric motor that drives two different pumps. Fracturing pump transport **108** may be arranged such that one pump is coupled to opposite ends of the dual shaft electric motor and avoids coupling the pumps in series. By avoiding coupling the pump in series, fracturing pump transport **108** may continue to operate when either one of the pumps fails or has been removed from fracturing pump transport **108**. Additionally, repairs to the pumps may be performed without disconnecting the system manifolds that connect fracturing pump transport **108** to other fracturing equipment within mobile fracturing system **103** and wellhead **101**. Fracturing pump transport **108** may be communicatively coupled with an AI system that is configured to detect predetermined states (e.g., states related to fluid end packing failure) that may occur at fracturing pump transport **108** during operation. Configuration of fracturing pump transport **108** is described in further detail in connection with FIGS. 2-11 below.

(38) Blender transport **110** may receive electric power fed through switch gear transport **112** to power a plurality of electric blenders. A plurality of prime movers may drive one or more pumps

that pump source fluid and blender additives (e.g., sand) into a blending tub, mix the source fluid and blender additives together to form fracturing fluid, and discharge the fracturing fluid to fracturing pump transport **108**. In one embodiment, the electric blender may be a dual configuration blender that comprises electric motors for the rotating machinery that are located on a single transport, which is described in more detail in U.S. Pat. No. 9,366,114, filed Apr. 6, 2012 by *Coli et al.* and entitled “Mobile, Modular, Electrically Powered System for use in Fracturing Underground Formations,” which is herein incorporated by reference in its entirety. In another embodiment, a plurality of enclosed mixer hoppers may be used to supply the proppants and additives into a plurality of blending tubs. Blender transport **110** may be communicatively coupled with an AI system that is configured to detect predetermined states (e.g., sand overflow, sand level, fluid backup, and the like) that may occur at blender transport **110** during operation. Configuration of blender transport **110** is described in further detail in connection with FIGS. 7-13 below.

(39) Data van **114** may be part of a control network system (e.g., AI system), where data van **114** acts as a control center configured to monitor and provide operating instructions to remotely operate blender transport **110**, power generation transport **102**, and fracturing pump transport **108** and/or other fracturing equipment within mobile fracturing system **103**. For example, data van **114** may implement the AI system that detects occurrence of predetermined states associated with one or more of blender transport **110**, power generation transport **102**, and fracturing pump transport **108** and/or other fracturing equipment within mobile fracturing system **103**. Data van **114** implementing the AI system may further perform predetermined operations or functions based on the detected predetermined state. In one embodiment, data van **114** may communicate via the control network system (e.g., AI system) with the variable frequency drives (VFDs) located within power generation transport **102** and/or switch gear transport **112** that operate and monitor the health of the electric motors used to drive the pumps on fracturing pump transports **108**. In one embodiment, data van **114** may communicate with the variety of fracturing equipment using a control network system that has a ring topology (or star topology). A ring topology may reduce the amount of control cabling used for fracturing operations and increase the capacity and speed of data transfers and communication.

(40) Other fracturing equipment shown in FIG. 1, such as fracturing liquid (e.g., water) tanks, chemical storage of chemical additives, hydration unit, sand conveyor, and sandbox storage are known by persons of ordinary skill in the art, and therefore are not discussed in further detail. In one or more embodiments of mobile fracturing system **103**, one or more of the other fracturing equipment shown in FIG. 1 may be configured to receive power generated from power generation transport **102**. The control network system for mobile fracturing system **103** may remotely synchronize and/or slave the electric blender of blender transport **110** with the electric motors of fracturing pump transports **108**. Unlike a conventional diesel-powered blender, the electric blenders may perform rate changes to the pump rate change mounted on fracturing pump transports **108**. In other words, if the pumps within fracturing pump transports **108** perform a rate change increase, the electric blender within blender transport **110** may also automatically compensate its rate and ancillary equipment, such as the sand conveyor, to accommodate the rate change. Manual commands from an operator may not be used to perform the rate change.

(41) Power generation transport **102** may be a part of mobile fracturing system **103** used at well site **101** as described in FIG. 1. Power generation transport **102** may be configured to be transportable to different well sites along with other equipment (e.g., fracturing pump transports) that is part of the mobile fracturing system **103** and may not be left behind after completing fracturing operations. Power generation transport **102** may include one or more transports (e.g., a power generation transport, inlet and exhaust transport, and the like) that are configured for minimizing operations for the mobilization and de-mobilization process. For example, power generation transport **102** may improve mobility by enabling a mobilization and de-mobilization time period of about 24 hours. Power generation transport **102** may have a single transport footprint, where the same

transport may be used in transportation and operational modes, and be configured as a 'self-sufficient' transport that carries all ancillary equipment for mobile electric power generation. Alternately, power generation transport **102** may have a multi transport footprint including a power generation transport and an inlet and exhaust transport. To provide electric power at one or more locations (e.g., well sites), power generation transport **102** may be designed to unitize and mobilize a gas turbine and a generator adapted to convert hydrocarbon fuel, such as natural gas, into electricity.

(42) Although not shown in FIG. 1, power generation transport **102** may include a variety of equipment for mobile electric power generation including a gas conditioning skid, a black start generator, a power source (e.g., gas turbine), a power source air inlet filter housing, a power source inlet plenum, a power source exhaust collector, an exhaust coupling member, a power source exhaust stack, a gearbox, a generator shaft, a generator, a generator air inlet filter housing, a generator ventilation outlet, a generator breaker, a transformer, a starter motor, and a control system. Other components on power generation transport **102** may include a turbine lube oil system, a fire suppression system, a generator lube oil system, and the like.

(43) In one embodiment, the power source may be a gas turbine. In another embodiment, power source may be another type of power source (e.g., diesel engine). The gas turbine may generate mechanical energy (e.g., rotation of a shaft) from a hydrocarbon fuel source, such as natural gas, liquefied natural gas, condensate, and/or other liquid fuels. For example, a shaft of the gas turbine may be connected to the gearbox and the generator such that the generator converts the supplied mechanical energy from the rotation of the shaft of the gas turbine to produce electric power. The gas turbine may be a commercially available gas turbine such as a General Electric NovaLT5 gas turbine, a Pratt and Whitney gas turbine, or any other similar gas turbine. The generator may be a commercially available generator such as a Brush generator, a WEG generator, or other similar generator configured to generate a compatible amount of electric power. For example, the combination of the gas turbine, the gearbox, and the generator within power generation transport **102** may generate electric power from a range of at least about 1 megawatt (MW) to about 36 MW (e.g., 5.6 MW or 32 MW). Other types of gas turbine/generator combinations with power ranges greater than about 36 MW or less than about 1 MW may also be used depending on the application requirement. In one embodiment, to increase mobility of power generation transport **102** and so that power generation transport **102** can be configured as a single transport, the gas turbine may be configured to fit within a dimension of about 14.5 feet long and about 4 feet in diameter and/or the generator may be configured to fit within a dimension of about 18 feet long and about 7 feet wide.

(44) As discussed above with reference to FIG. 1, the mobile fracturing system **103** includes a number of (mechanical and electrical) components that operate under harsh conditions. Careful monitoring of the operation of these components with the AI system disclosed herein can help keep the fracturing system **103** operational and can avoid failures, lost operation time, etc. The fracturing pump transport **108** of the hydraulic fracturing fleet **103** has mechanical components that can be monitored and controlled by the disclosed AI system of the present disclosure.

(45) To that end, discussion turns to FIGS. 2A and 2B, which are schematic diagrams of embodiments of fracturing pump transport **108** powered by power generation transport **102** as described in FIG. 1. Fracturing pump transport **108** may include prime mover **204** powering two separate pumps **202A** and **202B**. By combining a single prime mover **204** attached to two separate pumps **202A** and **202B** on a transport, a fracturing operation may reduce the amount of pump transports, prime movers, variable frequency drives (VFD's), ground iron, suction hoses, and/or manifold transports. Although FIGS. 2A and 2B illustrate that fracturing pump transport **108** supports a single prime mover **204** powering two separate pumps **202A** and **202B**, other embodiments of fracturing pump transport **108** may include a plurality of prime movers **204** that power respective one of pumps **202A** and **202B**.

(46) Fracturing pump transport may include trailer **210** having a "lay-down" design. Such a design

may provide mobility, improved safety, and enhanced ergonomics for crew members to perform routine maintenance and operations of the pumps since the “lay-down” arrangement positions pumps **202A** and **202B** lower to the ground as the main trailer beams are resting on the ground for operational mode. As shown in FIGS. 2A and 2B, “lay-down” trailer **210** has an upper section above the trailer axles that could hold or have mounted fracturing pump trailer power and control system **208**. Fracturing pump trailer power and control system **208** may comprise one or more electric drives, transformers, controls (e.g., a programmable logic controller (PLC) located on the fracturing pump transport **108**), and cables for connection to the drive power trailers and/or a separate electric pumper system. The electric drives may provide control, monitoring, and reliability functionality, such as preventing damage to a grounded or shorted prime mover **204** and/or preventing overheating of components (e.g., semiconductor chips) within the electric drives. The lower section of lay-down trailer **210**, which may be positioned lower than the trailer axles, may hold or have mounted prime mover **204** and pumps **202A** and **202B** attached on opposite sides of each other.

(47) In one embodiment, prime mover **204** may be a dual shaft electric motor that has a shaft that protrudes on opposite sides of the electric motor. The dual shaft electric motor may be any desired type of alternating current (AC) or direct current (DC) motor. In one embodiment, the dual shaft electric motor may be an induction motor. In another embodiment the dual shaft electric motor may be a permanent magnet motor. Other embodiments of prime mover **204** may include other electric motors that are configured to provide about 5,000 HP or more. For example, the dual shaft electric motor may deliver motor power in a range from about 1,500 HP to about 10,000 HP. Specific to some embodiments, the dual shaft electric motor may be about a 5,000 HP rated electric motor or about a 10,000 HP electric motor. Prime mover **204** may be driven by at least one variable frequency drive that is rated to a maximum of about 5,000 HP and may receive electric power generated from power generation transport **102**.

(48) As shown in FIGS. 2A and 2B, one side of prime mover **204** drives one pump **202A** and the opposite side of prime mover **204** drives a second pump **202B**. Pumps **202A** and **202B** are not configured in a series configuration in relation to prime mover **204**. In other words, prime mover **204** independently drives each pump **202A** and **202B** such that if one pump fails, it can be disconnected, and the other pump can continue to operate. Prime mover **204**, which could be a dual shaft electric motor, eliminates the use of diesel engines and transmissions. Moreover, using a dual shaft electric motor on a transport may prevent dissonance or feedback when transferring power to pumps **202A** and **202B**. In one embodiment, prime mover **204** may be configured to deliver at least about 5,000 HP distributed between the two pumps **202A** and **202B**. For instance, prime mover **204**, which may be a dual shaft electric motor, may provide about 2,500 HP to one of the pumps **202A** and about 2,500 HP to the other pump **202B** in order to deliver a total of about 5,000 HP. Other embodiments may have prime mover **204** deliver less than 5,000 HP or more than 5,000 HP. For example, prime mover **204** may deliver a total of about 3,000 HP by delivering about 1,500 HP to one of the pumps and about 1,500 HP to the other pump. Another example may have prime mover **204** deliver a total of about 10,000 HP by delivering about 5,000 HP to one of the pumps **202A** and about 5,000 HP to the other pump **202B**. Specifically, in one or more embodiments, prime mover **204** may operate at HP ratings of about 3,000 HP, 3,500 HP, 4,000 HP, 4,500 HP, 5,000 HP, 5,200 HP, 5,400 HP, 6,000 HP, 7,000 HP, 8,000 HP, 9,000 HP, and/or 10,000 HP.

(49) Fracturing pump transport **108** may reduce the footprint of fracturing equipment on a well-site by placing two pumps **202A** and **202B** on a single transport. Larger pumps may be coupled to a dual shaft electric motor that operates with larger horse power to produce additional equipment footprint reductions. In one embodiment, each of pumps **202A** and **202B** may be quintuplex pumps located on a single transport. Other embodiments may include other types of plunger style pumps, such as triplex pumps. Pumps **202A** and **202B** may each operate from a range of about 1,500 HP to about 5,000 HP. Specifically, in one or more embodiments, each of pumps **202A** and **202B** may

operate at HP ratings of about 1,500 HP, 1,750 HP, 2,000 HP, 2,250 HP, 2,500 HP, 2,600 HP, 2,700 HP, 3,000 HP, 3,500 HP, 4,000 HP, 4,500 HP, and/or 5,000 HP. Pumps **202A** and **202B** may not be configured in a series configuration where prime mover **204** drives a first pump **202A** and the first pump **202B** subsequently drives a second pump **202B**.

(50) Prime mover **204** and each of pumps **202A** and **202B** may be mounted on sub-assemblies configured to be isolated and allow for individual removal from fracturing pump transport **108**. In other words, prime mover **204** and each of pumps **202A** and **202B** can be removed for service and replaced without shutting down or compromising other portions of fracturing pump transport **108**. That is, prime mover **204** and pumps **202A** and **202B** may be connected to each other via couplings that are disconnected when removed from fracturing pump transport **108**. If prime mover **204** needs to be replaced or removed for repair, prime mover **204** sub-assembly may be detached from fracturing pump transport **108** without removing the two pumps **202A** and **202B** from fracturing pump transport **108**. Similarly, pump **202A** can be isolated from fracturing pump transport **108**, removed, and replaced by a new pump **202A**. If prime mover **204** and/or pumps **202A** and **202B** require service, an operator can isolate the different components from the fluid lines, and unplug, un-pin, and remove prime mover **204** and/or pumps **202A** and **202B** from fracturing pump transport **108**. Furthermore, each pump **202A/202B** sub-assembly may be detached and removed from fracturing pump transport **108** without removal of the other pump **202A/202B** and/or prime mover **204**. As such, fracturing pump transport **108** may not need to be disconnected from the manifold system and driven out of location at the well site. Instead, replacement prime mover **204** and/or pumps **202A** and **202B** may be placed back into the line and reconnected to fracturing pump transport **108**.

(51) To implement independent removal of the sub-assemblies, the two pumps **202A** and **202B** may be coupled to prime mover **204** using respective drive line assemblies **206**, each of which is adapted to provide local or remote operation to engage or dis-engage respective one of pumps **202A** and **202B** from prime mover **204**. Each drive line assembly **206** may comprise one or more couplings and a drive shaft. For example, drive line assembly **206** may comprise a fixed coupling that connects to one of pumps **202A** or **202B** and a corresponding keyed shaft **212**. Keyed shaft **212** may interconnect the fixed coupling to a corresponding splined toothed coupling **214** that is attached to prime mover **204**. To engage or disengage one or both pumps **202A** and **202B** from prime mover **204**, each spline toothed coupling **214** may include a splined sliding sleeve motor and pump coupling that provides motor shaft alignment and provides for a hydraulic fluid powered for connection and disconnection of the sliding sleeve motor and pump coupling. Other embodiments of the couplings may include torque tubes, air clutches, electro-magnetic clutches, hydraulic clutches, and/or other clutches and disconnects that have manual and/or remote operated disconnect devices.

(52) FIGS. 2A-2B illustrate that fracturing pump transport **108** includes engagement panel **220** for adjusting each spline toothed coupling **214** to engage and disengage pumps **202A** and **202B** from prime mover **204**. As an example, engagement panel **220** includes levers or switches that an operator manually operates to engage or disengage one or both pumps **202A** and **202B** from prime mover **204**. Additionally, or alternatively, to engage or disengage one or both pumps **202A** and **202B** from prime mover **204**, engagement panel **220** may include electronic controllers that receive instructions from remote locations, such as a monitoring station that is part of power and control system **208**, another location at the well site (e.g., AI system on data van **114**), and/or off-site via a network (e.g., the Internet). For example, if both pumps **202A** and **202B** are initially in an engaged position, in response to receiving a remote command, engagement panel **220** may trigger disengagement of one pump **202B** (so as to stop fluid pumping operation of pump **202B**) while the other pump **202A** remains in the engaged position (so as to allow pump **202A** to pump high-pressure fluid into wellhead **101**).

(53) FIGS. 2A-2B further illustrate that each of pumps **202A** and **202B** includes fluid end assembly

216 and power end assembly **218** that couples to a corresponding fluid end assembly **216**. Each power end assembly **218** generates torque to drive a corresponding fluid end assembly **216** (e.g., plungers) of each of pumps **202A** and **202B**. Power end assembly **218** may include a gear box including pinion gears and/or bull gears that rotate based on torque input from the drive shaft of drive line assembly **206** driven by prime mover **204**. Rotating the pinion gears causes the bull gears to rotate, which in turn causes rotation of a crankshaft within power end assembly **218** of pumps **202A** and **202B**. Rotation of the crankshaft then produces torque that moves plungers in fluid end assembly **216** of pumps **202A** and **202B** to pump and pressurize fracturing fluid into wellhead **101**. Pumps **202A** and **202B** are illustrated in greater detail in FIGS. **3A** and **3B**.

(54) In particular, FIG. **3A** illustrates an exposed view of pump **202** (e.g., pump **202A** or **202B**) provided on fracturing pump transport **108** in accordance with one or more embodiments. FIG. **3B** illustrates a cross-section view of fluid end assembly **216** of pump **202** provided on fracturing pump transport **108** in accordance with one or more embodiments. As shown in FIGS. **3A-3B**, pump **202** may be a reciprocating, positive displacement, horizontal single-acting pump in which rotational motion from a prime mover (e.g., prime mover **204**) at a given horsepower is applied to an input shaft flange. Power end assembly **218** of pump **202** may convert rotational motion into linear reciprocating motion of plunger **350**. In turn, plungers **350** of fluid end assembly **216** reciprocate in fluid end assembly **216** displacing a fixed volume of fluid with a suction stroke for incoming fluid from suction end **315**, and a power stroke for discharge fluid from discharge end **355**. Fluid may thus be compressed to discharge pressure and pushed through discharge end **355**.

(55) Power end assembly **218** may include multiple sub-assemblies including a crankshaft housing, a crosshead section, a spacer section, and a gearbox. As shown in FIGS. **3A-3B**, pump **202** may, for example, be a quintuplex pump in which five pony rods **305** of the crosshead section of power end assembly **218** are respectively coupled to five plungers **350** of fluid end assembly **216** via corresponding pony rod clamps **310**. Pony rods **305** may be actuated by the crankshaft sub-assembly of power end assembly **218** to generate linear reciprocating motion, and in turn, linearly reciprocate respective plungers **350** in and out of fluid end assembly **216** on the suction and discharge strokes to pump high-pressure fluid (e.g., mixture of sand and water for hydraulic fracturing) into wellhead **101**.

(56) Each plunger **350** may be configured to move in and out of a main cylinder of fluid end assembly **216** via a corresponding plunger bore **335**. The plunger **350** may be made of a lower carbon steel and may have a wear finish deposited and polished thereon to a smooth surface for proper sealing. Each plunger bore **335** may receive a corresponding plunger **350**, which is reciprocated in the plunger bore **335** to pump fluid through respective discharge ends **355**. The plunger end of each plunger bore **335** is sealed by packing in stuffing box **320**. The packing in stuffing box **320** for each plunger **350**/plunger bore **335** combination may include multiple components including a series of packing elements (e.g., packing ring **330**). Packing nut or gland **325** may be provided around an outer periphery of plunger **350** to compress the packing and to provide a high-pressure seal. A packing wiper seal received in a peripheral groove in packing nut **325** may seal stuffing box **320**. Lubrication ports may also be provided that permit lubricant to be pumped into stuffing box **320** to provide lubrication between plunger **350** and plunger bore **335**, between plunger **350** and packing nut **325**, and between plunger **350** and stuffing box **320**.

(57) As explained previously, the packing in stuffing box **320** may fail during operation due to a variety of reasons. For example, one or more of the series of packing elements (e.g., packing ring **330**) or packing nut **325** may fail during operation because of wear, lack of ample lubrication, poor lubrication, packing in stuffing box **320** being too tight or too loose around plunger **350**, packing nut **325** being too tight or too loose, and the like. The packing in stuffing box **320** or packing nut **325** may begin to fail. Seals in the packing nut **325** may become worn, and may washout or start washboarding. Proper lubrication of the plunger packing may be lost, increasing friction. Sand (or other additive) in the hydraulic fracking fluid mixture may get wedged in the packing and start to

cut away on the series of packing elements or on plunger **350** while it is linearly reciprocating with respect to fluid end assembly **216**. As a result of this friction, plunger **350** (or other components of fluid end assembly **216** and/or power end assembly **218**) may get hot, begin to smoke, and/or a high-pressure stream of the fluid and sand mixture may begin to escape out of the washout or cut formed in the packing in stuffing box **320**, packing nut **325**, or plunger **350**. The smooth surface of the plunger **350** may develop wear, pits, nicks, or burs that can further damage the stuffing box **320** and can lead to leaking. Scale, buildup, or cracks may also develop on the plunger **350**, compromising sealing.

(58) In order to monitor the current state of fluid end assembly **216** during operation, as shown in FIGS. 2A-2B and 3A-3B, fracturing pump transport **108** may include one or more sensors **205**. Sensor **205** can be exposed to an external environment of at least one mechanical component of the fracturing pump transport **108**, such as the fluid end assembly **216** of pump **202** or a portion thereof. Sensor **205** is configured to detect at least one external indicia of the operation of the at least one mechanical component. In general, sensor **205** can be an imaging sensor detecting visual characteristics, a thermal sensor detecting thermal characteristics, an acoustic sensor detecting sound, and a motion sensor detecting motion related to the operation of the at least one mechanical component. Proper operation of the mechanical component would be indicated by a consistent pattern of imaging, thermal, acoustic, and motion detections. A change in this consistent pattern would thereby equate to a potential issue with the operation of the mechanical component, such as indications of a failure, a leak, or the like.

(59) Overall, sensors **205** may include one or more of a plurality of types of sensors including acoustic or sound sensors (e.g., microphone and the like); optical, light or imaging sensors (e.g., charge coupled device (CCD) sensor, complimentary metal-oxide semiconductor (CMOS) sensor, electro-optical sensor, colorimeter, infrared sensor, thermal imaging sensor, and the like); flow or fluid velocity sensor (e.g., flow sensor and the like); environment or weather sensor; thermal, heat sensor; position, angle, displacement, distance, speed or acceleration sensor (e.g., laser rangefinder and the like); and the like. For example, sensors **205** may include an optical activity sensor, an optical sensor array, an accelerometer, a sound sensor, a barometric sensor, a proximity sensor, an ambient light sensor, a vibration sensor, a gyroscopic sensor, a compass, a barometer, a magnetometer, a thermistor sensor, an electrostatic sensor, a temperature sensor, a heat sensor, a thermometer, a light sensor, a differential light sensor, an opacity sensor, a scattering light sensor, a diffractional sensor, a refraction sensor, a reflection sensor, a polarization sensor, a phase sensor, a florescence sensor, a phosphorescence sensor, a pixel array, a micro pixel array, a rotation sensor, a velocity sensor, an inclinometer, a pyranometer, a momentum sensor, a pump pressure sensors, and a wave radar probe.

(60) In the example embodiment shown in FIGS. 2A-2B and 3A-3B, three sensors **205** are illustrated as being positioned on either side of and on top of fluid end assembly **216** of each pump **202A** and **202B** of each frac pump transport **108**. However, this may not necessarily be the case. The number, type, position, angle, and other characteristics of sensors **205** are not intended to be limiting to that shown in the drawings, and may be determined based on the use case (e.g., what equipment, apparatuses, or components of the hydraulic fracturing system **103** are being monitored), and the types of predetermined states (e.g., what are the relevant operational states of the monitored equipment or component) that are being monitored by sensors **205**.

(61) As one non-limiting example, sensors **205** may be imaging sensors (e.g., digital camera) that are positioned to face plungers **350** of fluid end assembly **216** of each pump **202A** and **202B** of each transport **108** to detect fluid end packing states. For example, as shown in FIGS. 3A-3B, one or more sensors **205** (e.g., cameras) may be positioned on each pump **202A** or **202B** of each frac pump transport **108** at fluid end assembly **216** to provide a video feed of plungers **350** that are linearly reciprocating with respect to the main cylinder of fluid end assembly **216**, and more particularly, with respect to the stationary packing nut **325** and packing in stuffing box **320**

provided on the main cylinder of fluid end assembly **216**. One (or more) camera may be positioned for each plunger **350** so that its frame of view includes one plunger's **350** reciprocating motion with respect to the stationary packing nut **325** and stuffing box **320** provided on the main cylinder of fluid end assembly **216**.

(62) Alternately, one (or more) cameras may be positioned for each pump **202** so that its frame of view includes all plungers **350** in reciprocating motion with respect to corresponding stationary packing nuts **325** and packings in stuffing boxes **320** in corresponding plunger bores **335** on the main cylinder of pump **202**. For example, as shown in FIG. 4, sensor **205** (e.g., camera) may be positioned relative to pump **202** so that image data **410** output by sensor **205** may show each pony rod **305**, each pony rod clamp **310**, each plunger **350**, and each packing nut **325** of pump **202**. As another example, as shown in FIG. 5, sensor **205** (e.g., camera) may be positioned at an elevated position relative to pump **202** so that image data **510** output by sensor **205** may show spacer section **520** of power end assembly **218**, as well as show each pony rod **305**, each pony rod clamp **310**, each plunger **350**, and each packing nut **325** of pump **202**.

(63) With respect to the sensors **205** being imaging sensors to detect the fluid end packing states, the imaging sensors **205** image the reciprocation of the plungers **350** relative to the stuffing boxes **320**, packing gland **325**, packing ring **330**, etc. During the repeated reciprocation, the imaging sensors **205** can image the external features and environment around these components during normal operation. When a leak or failure begins, the consistent nature of the imaging will change as the imaging sensor **205** detects smoke emanating from worn or over-heated elements, burst of high-pressure fluid leaking from the elements, mist, spray, or foam caused by leak of the high-pressure fluid from the components around plungers **350**, and the like.

(64) As another non-limiting example, one or more of sensors **205** may be a thermal imaging sensor that is configured to detect thermal characteristics (e.g., temperature) of plungers **350** and surrounding equipment (e.g., packing nut **325**, stuffing box **320**, main cylinder of fluid end assembly **216**, and the like). Performing thermal imaging, sensors **205** image the reciprocation of the plungers **350** relative to the stuffing boxes **320**, packing gland **325**, packing ring **330**, etc. During the repeated reciprocation, the thermal imaging sensors **205** can image the thermal signature and environment around these components during normal operation. When a leak or failure begins, the consistent nature of the thermal signature will change as the thermal imaging sensor **205** detects hot smoke emanating from worn or over-heated elements, burst of high-pressure fluid leaking from the elements, increase in temperature of plunger **350**, and surrounding equipment, and the like.

(65) As noted, the sensors **205** can include acoustic sensors (e.g., microphone and the like) to monitor the acoustic environment of the plungers **350** and surrounding equipment (e.g., packing nut **325**, stuffing box **320**, main cylinder of fluid end assembly **216**, and the like). The acoustic sensors **205** can detect the acoustic signature of the repeated reciprocation of these components during normal operation. When a leak or failure begins, the consistent nature of the acoustic signature will change as the acoustic sensor **205** detects predetermined (learned) acoustic signatures representing burst of high-pressure fluid leaking from the elements, mist, spray, or foam caused by leak of the high-pressure fluid from the components around plungers **350**, and the like.

(66) As a vibration sensor detecting vibration, the sensor **205** can monitor vibrations in the environment of the plungers **350** and surrounding equipment (e.g., packing nut **325**, stuffing box **320**, main cylinder of fluid end assembly **216**, and the like). The vibration sensors **205** can detect the vibrational signature of the repeated reciprocation of these components during normal operation. When a leak or failure begins, the consistent nature of the vibrational signature will change as the vibration sensor **205** detects predetermined (learned) vibrational signatures representing overheated components, burst of high-pressure fluid leaking from the elements, mist, spray, or foam caused by leak of the high-pressure fluid from the components around plungers **350**, and the like.

(67) As yet another non-limiting example, one or more sensors **250** may include a combination of sensors like imaging sensors, thermal imaging sensors, audio sensors, flow sensors, and the like for each piece of equipment of hydraulic fracturing system **103** being monitored by the AI system.

(68) Returning to FIGS. 2A-2B, the one or more sensors **205** may be configured to capture data (e.g., still-image data, moving-image data, audio data, thermal still- or moving-image data, other sensor data, and the like) of plungers **350** of pump **202** during operation and transmit the generated sensor data to fracturing pump trailer power and control system **208**, data van **114** (e.g., implementing AI system), and/or to another remote-control network system. In one embodiment, the sensor data may be generated and transmitted continuously or at pre-determined intervals, based on an operation state of pump **202**.

(69) By monitoring data (e.g., image data **410** or **510**) generated by one or more sensors **205**, predetermined states associated with pump **202** can be detected by, e.g., an operator monitoring the data in real-time. However, as mentioned previously, in the event of a fluid end packing failure state where high-pressure fluid stream starts leaking out of fluid end assembly **216**, if the pump operator does not shut down the pump in a very rapid manner (e.g., within order of seconds), the high-pressure sand (or other additive) laden hydraulic fracturing fluid can cut a groove into the packing bore of the fluid end of the pump, and make the fluid end un-useable without costly and sometimes unreliable repair. It is desirable to automatically and dynamically detect predetermined states based on sensor data associated with one or more transports of the hydraulic fracturing system **103** and automatically perform predetermined operations or functions based on the detected predetermined states, in order to avoid damage to hydraulic fracturing equipment and minimize costly downtime.

(70) FIG. 6 is a block diagram of an embodiment of AI system **600** that monitors, controls, and communicates with one or more control systems associated with one or more transports (e.g., power generation transport **102**, fracturing pump transport **108**, blender transport **110**, switch gear transport **112**, and the like) of hydraulic fracturing system **103**. One use case for AI system **600** may be to detect occurrence of one or more predetermined states (e.g., overheating state, smoking state, fluid end packing failure (leaking) state, and the like) for fluid end assemblies **216** (e.g., apparatuses) of one or more frac pump transports **108** as described above in connection with FIGS. 2A-5. Other use cases for AI system **600** may be to detect occurrence of one or more predetermined states for other hydraulic fracturing equipment, components, or apparatuses on fracturing pump transport **108**, or on one or more of the other transports (e.g., blender transport **110**, power generation transport **102**, and the like) based on corresponding sensor data generated by corresponding sensors **205** and configured for the particular use case (or the particular apparatuses).

(71) AI system **600** may be implemented as a control network system that monitors, controls, and communicates with one or more control systems of hydraulic fracturing system **103**. In one embodiment, AI system **600** may include control center **605** that may be implemented on data van **114**. Alternately, or additionally, control center **605** may be implemented in a distributed manner on one or more control systems associated with one or more transports (e.g., power generation transport **102**, fracturing pump transport **108**, blender transport **110**, switch gear transport **112**, data van **114**, and the like) of hydraulic fracturing system **103**. In yet another embodiment, control center **605** may be implemented on a computer system that is separate from data van **114** and the control systems of the one or more transports of fracturing system **103**. For example, control center **605** may be implemented on a remote computer system (e.g., remote well site computer system) that is communicatively coupled to data van **114** and the control systems of the one or more transports of fracturing system **103** via a network (e.g., the Internet). As another example, control center **605** may be implemented on a cloud-based computing system.

(72) AI system **600** may be implemented using any suitable network topology (e.g., ring topology, star topology, mesh topology, bus topology, and the like) that interconnects control center **605**, blender transport(s) **110**, chemical additive unit **610**, hydration unit **615**, fracturing pump

transport(s) **108**, sensors **205**, and power generation transport **102**. In one embodiment, AI system **600** may be implemented as a ring topology network that may reduce the amount of control cabling used for fracturing operations and increase the capacity and speed of data transfers and communication. Additionally, the ring topology may allow for two-way communication and control by control center **605** for equipment connected to control center **605**. For example, control center **605** may be able to monitor and control other fracturing equipment and third-party equipment when added to AI system **600**, and for multiple pieces of equipment to communicate with each other. In one embodiment, AI system **600** may be a network, such as an Ethernet network that connects and communicates with individual control systems for each fracturing transport (e.g., one or more of power generation transport **102**, fracturing pump transport **108**, blender transport **110**, switch gear transport **112**, data van **114** and the like). Control center **605** may be configured to monitor, control, and provide operating instructions to different fracturing equipment on the one or more transports of fracturing system **103**. For example, control center **605** may communicate with engagement panel **220** located within fracturing pump transport **108** to operate prime mover **204** and one or both pumps **202A** and **202B**.

(73) As shown in FIG. 6, control center **605** of AI system **600** may include storage device **620**, memory **625**, one or more AI models **630**, control unit **640**, and input/output devices **650**. Storage device **620** and memory **625** may store and/or retrieve data, which can include sensor data from one or more sensors **205**, AI model data for one or more use cases (or corresponding apparatuses), data associated with power generation transport **102**, data associated with chemical additive unit **610**, data associated with hydration unit **615**, data associated with power generation transport **102**, data associated with blender transport **110**, data describing input/output devices **650**, and/or metadata associated with the data. Memory **625** of control center **605** may be implemented using any suitable (volatile and/or non-volatile) memory technology to store program instructions and data associated with control center **605**.

(74) Control center **605** may also include input/output devices **650**. For example, input/output devices **650** may include at least one of the following: (i) one or more input devices that interact with or send data to one or more components of control center **605** (e.g., mouse, keyboards, wireless Bluetooth® peripheral, etc.); (ii) one or more output devices that provide output from one or more components of control center **605** (e.g., internal or built-in monitors, printers, haptic feedback devices/systems, etc.); or (iii) one or more storage devices that store data in addition to memory **625** and storage device **620**. (Bluetooth® is a registered trademark of Bluetooth SIG, Inc, Washington, USA). Input/output devices **650** may combine different devices into a single hardware component that can be used both as an input and output device (e.g., a touchscreen, and the like).

(75) Control unit **640** may include CPUs, GPUs, digital signal processors (DSPs), and/or other types of integrated circuits (ICs). Control unit **640** may also contain circuits such as memory, processors, application specific integrated circuits, and other storage and processing circuitry. Control unit **640** may be configured to communicate with other components of AI system **600** such as sensors **205**, power generation transport **102**, blender transport **110**, fracturing pump transport **108**, and the like. Control unit **640** may enable AI model **630** to detect predetermined states for one or more use cases (or apparatuses) and perform predetermined operations or functions based on the detected predetermined states with respect to equipment on one or more hydraulic fracturing transports of hydraulic fracturing system **103**.

(76) AI model **630** may be a mathematical model (e.g., machine or deep learning model) for making data-driven predictions or decisions (e.g., predictions, classifications, diagnosis, and the like). Known data (e.g., sensor data) and known responses (e.g., particular image known to be corresponding to a particular predetermined state) may be used to create AI model **630** for a particular use case (or for corresponding apparatuses). AI model **630** may then be deployed on control center **605**, and new data (e.g., live or real-time data from sensors **205** (e.g., image sensors, thermal imaging sensors, and the like) showing plungers **350** of fluid end assembly **216** in

operation) may be fed into AI model **630**, which may then produce responses (e.g., predictions, classifications, diagnosis) to the new input data, so as to detect one of plural predetermined states AI model **630** is designed to detect for a particular use case.

(77) For example, in the fluid end packing failure use case, AI model **630** may be trained to detect any of plural predetermined states including, a ‘normal’ operation state in which AI model **630** detects that a particular fluid end assembly **216** is operating normally during the hydraulic fracturing operation, a ‘notify’ operation state in which AI model **630** detects that components of the particular fluid end assembly **216** are emanating smoke and are overheated, and a ‘auto-shutdown’ operation state in which AI model **630** detects that the particular fluid end assembly **216** is leaking high-pressure fracturing fluid during the hydraulic fracturing operation.

(78) For example, the sensor **205** as an imaging sensor can detect visual characteristics of the operation of plunger **350** and stuffing box **320** as plunger **350** repeatedly reciprocates during normal operation, as noted previously. Leaking of high-pressure fracturing fluid from gland **325**, smoking due to wear of stuffing box **320**, and the like would produce visual indicia of the issue. The AI model **630** determines the state of fluid end assembly **216** based on this visual indicia and outputs the appropriate notification or shut-down operation state.

(79) For example, the sensor **205** as a thermal sensor can detect thermal characteristics of the operation of plunger **350** and stuffing box **320** as plunger **350** repeatedly reciprocates during normal operation, as noted previously. Leaking of high-pressure fracturing fluid from the gland **325**, smoking due to wear of stuffing box **320**, and the like would produce thermal indicia of the issue. The AI model **630** determines the state of fluid end assembly **216** based on this thermal indicia and outputs the appropriate notification or shut-down operation state.

(80) As will be appreciated, the sensor **205** as an acoustic sensor, a vibration sensor, or other sensor disclosed herein can detect the external, environmental characteristics of the operation of plunger **350** and stuffing box **320** as plunger **350** repeatedly reciprocates during normal operation, as noted previously. Failure(s) would produce indicia of the issue, and the AI model **630** determines the state of fluid end assembly **216** based on this indicia and outputs the appropriate notification or shut-down operation state.

(81) For each detected operation state for a particular use case, control center **605** may be configured to perform a corresponding predetermined operation or function. The predetermined operation performed by control center **605** may be predefined by a user for each predetermined state of the use case. For example, in the fluid end packing failure use case, when AI model **630** determines that fluid end assembly **216** is in the ‘normal’ operation state, control center **605** may be configured to notify an operator (e.g., by displaying a notification on a display device) of the ‘normal’ state and allow normal operations of fracturing equipment on fracturing pump transport **108**. When AI model **630** detects that fluid end assembly **216** is in the ‘notify’ operation state, control center **605** may be configured to notify an operator of the ‘notify’ state and display (e.g., by displaying a notification on a display device) information indicating the ‘abnormal’ behavior of fluid end assembly **216** during operation. When AI model **630** detects that fluid end assembly **216** is in the ‘auto-shutdown’ operation state, control center **605** may be configured to automatically shut down the pump (e.g., one of pumps **202A** and **202B** in FIGS. 2A-2B) corresponding to fluid end assembly **216** that is detected as leaking high-pressure fluid. The above described number and characteristics of the predetermined states and corresponding predetermined operations for a given use case are for illustration only. AI model **630** may instead be trained to detect other additional or alternative predetermined states and control center **605** may be configured to perform other additional or alternative corresponding operations for the fluid end packing failure use case (or for other use cases of AI system **600**).

(82) For each use case (or corresponding apparatuses), AI system **600** may include one or more trained and deployed AI models **630**. Each AI model **630** may also correspond to one or more use cases. In addition, each AI model **630** may be trained with and may utilize one or more types of

sensor data (e.g., image sensor, thermal imaging sensor, laser, infrared, audio sensor, and the like) for detecting predetermined states of one or more corresponding use cases. For each use case, corresponding sensors **205** (e.g., sensors **1, 2, . . . N**) may be configured to provide corresponding sensor data to control center **605**. AI models may thus be deployed anywhere and for any piece of equipment (e.g., component or apparatus) of hydraulic fracturing system **103** whose operation is to be monitored and automatically controlled based on corresponding predetermined states. AI models for each use case may be created at control center **605** or may be created at a back-office prior to subsequent deployment on AI system **600**. AI models **630** may be machine or deep learning models that rely on automated learning over time.

(83) FIG. 7 is dataflow graph **700** illustrating an exemplary technique for creating an AI model that recognizes patterns (e.g., detect predetermined states based on input image data) and learns over time, in accordance with one or more embodiments. Known data **705** is used to build an AI model (e.g., machine or deep learning model) for making data-driven decisions (e.g., predictions, classification, diagnosis). Known data **705** (e.g., known sensor data) may include known features and known responses used to train, validate and test a model that is then deployed as generalized model **730** (e.g., AI model **630**) on AI system **600** to detect predetermined states for a given use case related to hydraulic fracturing system **103**. For example, in the fluid end packing failure use case, known data **705** may include image data (e.g., images shown in FIGS. 4 and 5; known features) and information regarding a corresponding predetermined state (e.g., ‘normal’ state, ‘auto-shutdown’ state, ‘notify’ state, and the like; known responses) for each piece of known data. In another example, known data **705** can include other sensor data such as audio data, thermal imaging data, and the like, and information regarding a corresponding predetermined state each piece of known data may correspond to. Known data **705** can include multiple types or streams of sensor data. That is, an AI model may be trained based on multiple types of sensor data (e.g., audio, video, and thermal imaging data).

(84) The technique shown in FIG. 7 may be performed for each use case of AI system **600** for which an AI model is being created and deployed. Thus, for example, the technique shown in FIG. 7 may be performed for creating and deploying an AI model for the fluid end packing failure use case based on a first set of known data **705** specific to this use case, and the technique shown in FIG. 7 may be performed again for creating and deploying another AI model for a different use case (e.g., blender or hopper use case) based on a second set of known data **705** specific to the use case.

(85) Control center **605** (or another computer system) may subject known data **705** to pre-processing to clean and prepare the data to make it suitable for consumption by model training algorithms. An algorithm is a mathematical function that performs certain predetermined analysis on data and discovers insights from the data. The algorithm feeds an AI model which looks at characteristics of a process (e.g., use case having predetermined states) being examined and predicts how a system will behave with different variables. For example, the data (e.g., sensor **205** data) may be visualized and analyzed to run sanity checks to validate the quality of the data and to understand the data. Known data **705** may include variables and features that are attributes of examples that can be used to identify patterns to predict a target answer. Known data **705** may further include data that is labeled as a target (correct answer) associated with the attributes of the examples that the algorithm is intended to learn from and predict based on. After training, generalized model **730** is deployed and designed to predict the answer on data for which the target answer is not known. Control center **605** may further subject known data **705** to feature processing to make the features and variables of the dataset more meaningful. Feature processing may include forming more generalizable data points to learn from to provide significant improvements to the predictive, classification or diagnostic model. Common types of feature processing performed on the data may include replacing missing or invalid data, forming Cartesian products of one variable with another, non-linear transformations, creating domain-specific features, and creating variable-

specific features. In one embodiment, known data **705** may include a representation of each class (e.g., predetermined state) the corresponding model is designed to predict on. In other words, in the fluid end packing failure use case, known data **705** may include images for each of the three states (e.g., 'normal' state, 'auto-shutdown' state, 'notify' state) it is being trained to predict on.

(86) After pre-processing, control center **605** may enable (in hardware, software, or both) splitting known data **705** into training dataset **710**, validation dataset **715**, and test dataset **720**, in order to evaluate the model to estimate the quality of its pattern generalization for data the model has not been trained on. That is, since future data instances have unknown target values that cannot be used for checking accuracy of predictions of the model, some of the data from known data **705** for which we already know the answer of the prediction target for each row is used to evaluate the accuracy of the model (and underlying algorithm). For example, 70-80% of known data **705** may be used for training and 20-30% may be used for validation and testing. Further, control center **605** may use data in training dataset **710** to train models to see patterns and use data in test dataset **720** to evaluate predictive quality of the trained model. In one embodiment, control center **605** may further split training data into training dataset **710** and validation dataset **715**.

(87) Control center **605** may then use training dataset **710** to train different models by applying different classifiers and algorithms. Examples of machine learning and deep learning algorithms that may be used to train the model may include a naïve Bayes Classifier Algorithm, K Means Clustering Algorithm, Support Vector Machine Algorithm, Linear Regression, Logistic Regression, any kind of Neural Network, Random Forests, Decision Trees, Nearest Neighbors, or Gaussian process. That is, in order to find the most suitable classifier or algorithm for the problem to be solved by the model, control center **605** may use training dataset **710** to train multiple candidate algorithms and use validation dataset **715** to compare their performances and decide which candidate algorithm to select. Control center **605** may then create estimated model **725** based on the selected algorithm. Once estimated model **725** is created, control center **605** may use test dataset **720** to obtain performance characteristics such as accuracy, sensitivity, specificity, F-measure and the like of the selected algorithm to create generalized model **730**. Thus, validation dataset **715** may function as a hybrid dataset that is used as training data by testing, but neither as part of the low-level training nor as part of the final testing.

(88) Control center **605** may further tune hyperparameters or training parameters to improve predictive performance of generalized model **730**. In one embodiment, the candidate algorithms may be supervised, unsupervised or reinforcement machine learning algorithms depending on the characteristics of incoming data. Further, before deploying generalized model **730** in production (e.g., on AI system **600**), control center **605** may take steps to improve performance and accuracy of the model by improving model fit by determining whether the predictive model is overfitting or underfitting training dataset **710** by looking at prediction error on training dataset **710** and validation dataset **715**. The model is overfitting training dataset **710** when accuracy of training dataset **710** is higher than validation dataset **715**. Conversely, the model is underfitting training dataset **710** when the model performs poorly on training dataset **710**. Control center **605** may indicate corrective action to achieve balanced model fit before generalized model **730** is deployed. Once generalized model **730** that performs well is deployed, the model may be used to detect predetermined states (e.g., make predictions, classifications, diagnosis) on test field dataset **735** (i.e., live or real-time incoming data from sensors **205** during hydraulic fracturing operation) to generate predicted responses **740** (i.e., perform a prediction operation, diagnostic operation or classification operation on field data to detect one of a plurality of predetermined states for the incoming data). In one embodiment, control center **605** may make batch predictions using generalized model **730** to generate a prediction (e.g., predetermined state) for a set of observations (e.g., image data corresponding to a plurality of images) all at once based on a certain percentage of the observations meeting a threshold. Alternately, control center **605** may generate predictions on a one-by-one basis in real-time for each example (e.g., image data corresponding to each incoming

image) independent of the other examples based on incoming data.

(89) In order to keep generalized model **730** predicting accurately, control center **605** may continuously monitor incoming data and indicate that the model be updated (retrained) **745** based on new incoming data if control center **605** determines that data distribution of data on which AI model **630** is making predictions has deviated significantly from data distribution of original training dataset **710**. Alternately, control center **605** may simply indicate that the model be retrained **745** periodically (e.g., daily, weekly, or monthly) regardless of whether the data distribution has changed. Data distribution may change for a variety of reasons. Generalized model **730** is created based on data available at the time the model was created and deployed. When generalized model **730** was originally developed and deployed in AI system **600**, control center **605** may have made several assumptions relating to AI system **600** (e.g., data pipeline processes) that may change over time and no longer hold true. For example, data distribution may change because of changing infrastructure of AI system **600**, environmental changes, and new observations made (e.g., new features related to a given predetermined state of a given use case) that didn't exist at the time of model creation. As a result, data distribution of incoming data may deviate significantly from known data **705** that was used to train generalized model **730**, thereby lowering accuracy and efficiency of generalized model **730** over time and requiring updating and retraining of estimated model **725** based on the new incoming data.

(90) As explained above, control center **605** may include multiple AI models **630** that have been trained and deployed to handle multiple different use cases for automating predetermined state detection, and performance of corresponding predetermined operations or functions by control unit **640**. Control center **605** may be configured so that once an AI model **630** has been trained and deployed on control center **605**, an operator may be able to set predetermined operations that are to be performed when one of the predetermined states corresponding to the trained and deployed AI model **630** is detected in real-time. For example, input/output device **650** may provide an operator with a user interface to set and configure respective predetermined operations that are to be performed when each of the predetermined states of the AI model for the particular use case is detected.

(91) Thus, in the example described above of the fluid end packing failure use case, the user interface of input/output device **650** may enable the operator to selectively configure predetermined operations for each of the predetermined states (e.g., 'normal' state, 'auto-shutdown' state, 'notify' state). Further, the user interface of input/output device **650** may also enable the operator to change or reconfigure the predetermined operations for the predetermined states selectively. For example, if the underlying AI model **630** is updated to include a new predetermined state, the user interface of input/output device **650** may also enable the operator to set corresponding one or more predetermined operations for the new defined predetermined state the underlying AI model **630** is now capable of detecting.

(92) Thus, for example, in case of the 'notify' state, the user interface of input/output device **650** may enable the operator to configure a predetermined function in which control unit **640** may control input/output device **650** (or another display device) to display a notification to an operator indicating that the corresponding pump is overheating. In this case, the displayed notification may also include additional information like the underlying image data (e.g., an image showing an overheated and/or smoking plunger **350**) based on which the 'notify' state was predicted by the AI model, information identifying the particular fluid end assembly **216**, the particular pump **202A** or **202B**, and the particular fracturing pump transport **108** out of a plurality of fracturing pump transports **108** corresponding to the notification.

(93) For example, as shown in FIG. 1, well site **100** may include multiple fracturing pump transports **108**, each transport containing multiple pumps **202**. When one of the pumps of one of the transports **108** begins to fail (e.g., as indicated by it smoking or overheating) control unit **640** may identify the particular pump based on, for example, identification information of corresponding

sensor data indicating a corresponding sensor **205** positioned on the corresponding pump which is smoking, and control unit **640** may display this identifying information as part of the notification in response to detection of the ‘notify’ state of the pump. For example, control unit **640** may control input/output device **650** to display a map of well site **100** which shows the various transports **108** and pumps **202**, and further distinctively display the identified pump **202** on the identified transport **108** for which the notification instruction is being issued. Presentation of such information may enable an operator to easily and quickly ascertain the piece of equipment in a multi-transport fracturing fleet **103** that is potentially malfunctioning and/or beginning to malfunction, and act accordingly in a timely and highly efficient manner. For example, the notification displayed on input/output device **650** may instruct an operator to physically inspect the identified piece of equipment to determine next steps, thereby preventing potential irreparable damage to the equipment (e.g., fluid end assembly **216**).

(94) As another example, in case of the ‘auto-shutdown’ state, the user interface of input/output device **650** may enable the operator to configure a predetermined operation in which control unit **640** may control engagement panel **220** (or control system **208**) of the particular fracturing pump transport **108** whose pump **202** is detected to be in the ‘auto-shutdown’ state and cause the engagement panel **220** (or control system **208**) of the particular transport **108** to automatically shut down the particular pump (e.g., one of pump **202A** and **202B**) that is detected to be leaking by the underlying AI model **630**.

(95) In this case, the control instruction from control unit **640** may specifically identify the particular pump **202** and the particular transport **108** where the pump is located based on, for example, identification information of corresponding sensor data indicating a corresponding sensor **205** positioned on the corresponding pump which is leaking high-pressure fluid, and control unit **640** may utilize this identifying information to control a corresponding engagement panel **220** to control a corresponding pump. For example, upon receiving the instruction to shutdown pump **202A** from control unit **640**, engagement panel **220** may trigger disengagement of pump **202A** while pump **202B** remains in the engaged position and continues to operate. As a result, a particular pump **202** of a particular transport **108** of fracturing system **103** can be shutdown automatically based on state detection by a corresponding deployed AI model, thereby preventing potential irreparable damage to equipment and reducing downtime and costly repairs or replacement of equipment, component, or apparatus.

(96) While the predetermined operations have been described above on context of predetermined states of the fluid end packing failure use case, similar predetermined operations can be performed by control unit **640** to control one or more units of AI system **600** based on preconfigured operations defined by a user corresponding to respective predetermined states of one or more other use cases. Thus, control unit **640** may similarly monitor, control, and communicate with control system **603** of power generation transport **102** to control turbine **601** and/or generator **602** based on corresponding predetermined states detected by any underlying AI model **630** deployed on control center **605** and based on corresponding sensor data detected by corresponding sensors **205** in real-time during operation and including identification information identifying which equipment the sensor data corresponds to. Similarly, control unit **640** may monitor, control, and communicate with control system **660** of blender transport **110** to control blender **665** and/or hopper **670** based on corresponding predetermined states detected by any underlying AI model **630** deployed on control center **605** and based on corresponding sensor data detected by corresponding sensors **205** in real-time during operation and including identification information identifying which equipment the sensor data corresponds to. Control center **605** may similarly automatically control other equipment (e.g., chemical additive unit **610**, hydration unit **615**) of fracturing system **103** based on any corresponding detected predetermined states, any underlying created and deployed AI models, and any corresponding incoming sensor data from corresponding sensors **205**.

(97) In addition to the fracturing pump transport **108**, the blender transport **110** of the hydraulic

fracturing fleet **103** also has mechanical components that can be monitored and controlled by the disclosed AI system of the present disclosure. To that end, discussion turns to use cases for AI system **600** to monitor equipment on blender transport **110**, which are illustrated below in connection with FIGS. **8A-8B**, and **9A-9B**.

(98) FIGS. **8A** and **8B** are schematic diagrams of an embodiment of blender transport **110** that includes electric blender **806**. FIG. **8A** illustrates a top-down view of blender transport **110** and FIG. **8B** illustrates a side-profile view of blender transport **110**. Blender transport **110** may be powered by power generation transport **102** as described in FIG. **1**. Electric blender **806** may be a dual configuration blender with a blending capacity of about 240 bpm. The dual configuration blender may comprise electric motors for all rotating machinery and may be mounted on a single transport. The dual configuration blender may have two separate blending units that are configured to be independent and redundant. For example, any one or both the blending units may receive a source fluid via inlet manifolds of the blending units. The source fluid may originate from the same source or different sources. The source fluid may subsequently be blended by any one or both of the blending tub and subsequently discharged out of any one or both outlet manifolds of the blending units. Other embodiments of blender transport **110** may be a single configuration blender that includes a single blending unit.

(99) FIGS. **8A** and **8B** illustrate a “lay-down” trailer **802** design that provides mobility and improves ergonomics for the crew members that perform routine maintenance and operations of electric blender **806** as the “lay-down” design positions the blender tubs, pumps and piping lower to the ground level and the main trailer beams are resting on the ground for operational mode. Similar to “lay-down” trailer **210**, “lay-down” trailer **802** may comprise an upper section above the trailer axles and a lower section below the trailer axles. In one embodiment, electric blender **806** and associated equipment on the trailer may be controlled and monitored remotely via AI system **600**. As shown in FIGS. **8A** and **8B**, blender control system **804** that comprises a PLC, transformers and one or more variable frequency drives are mounted on upper section of blender transport **110**. To provide remote control and monitoring functions, control center **605** of AI system **600** may interface and communicate with the PLC (e.g., provide operating instructions), and the PLC may subsequently control one or more variable frequency drives mounted on blender trailer **110** to drive one or more electric motors of the blender.

(100) Operating blender transport **110** remotely may eliminate equipment operators from being exposed to hazardous environment and avoiding potential exposure to concentrated chemicals, silica dust, and rotating machinery. For example, a conventional blender transport typically includes a station for an operator to manually operate the blender. By remotely controlling using the control network and blender control system **804**, the station may be removed from the blender transport **110**. Recall that a data van may act as a hub to provide the remote control and monitoring functions and instructions to blender control system **804**.

(101) FIGS. **9A** and **9B** are schematic diagrams of an embodiment of blender transport **110** that includes electric blender **902** with enclosed mixer hoppers **904**. FIG. **9A** illustrates a top-down view of blender transport **110** and FIG. **9B** illustrates a side-profile view of blender transport **110**. Electric blender **902** is substantially similar to electric blender **806** except that electric blender **902** uses enclosed mixer hoppers **904** to add proppants and additives to the blending tub.

(102) FIGS. **9A** and **9B** illustrate that electric blender **902** is a dual configuration blender that includes two enclosed mixer hoppers **904** powered by two electric motors, where each of the electric motors may operate an enclosed mixer hopper **904**. Blenders that comprises open hoppers and augers typically have the proppants (e.g., sand) and/or additives exposed to the weather elements. In situations where precipitation occurs at the well site, operators may cover the open hoppers and augers with drapes, tarps, and/or other coverings to prevent the precipitation from contaminating the proppants and/or additives. Enclosed mixer hopper **904** replaces the open hopper and augers typically included in a blender (e.g., electric blender **806** in FIGS. **8A** and **8B**) with

enclosed mixer hoppers **904** (FIGS. **9A** and **9B**). By replacing the open hopper and augers with enclosed mixer hoppers **904** blender transport **110** may have the advantages of dust free volumetric proppant measurement, dust free mixing of proppant and additives, moderate the transport of proppants, perform accurate volumetric measurements, increase proppant transport efficiency with low slip, prevent proppant packing from vibration, produce a consistent volume independent of angle of repose, and meter and blend wet sand. Other advantages include the removal of gearboxes and increasing safety for operators with the enclosed drum.

(103) As shown in FIGS. **8A-8B** and **9A-9B**, blender transport **110** may also include one or more sensors **205**. Again, sensor **205** can be exposed to an external environment of at least one mechanical component of the blender transport **110**, such as blender, hopper, auger, or the like. Sensor **205** is configured to detect at least one external indicia of the operation of the at least one mechanical component. In general, sensor **205** can be an imaging sensor detecting visual characteristics, a thermal sensor detecting thermal characteristics, an acoustic sensor detecting sound, and a motion sensor detecting motion related to the operation of the at least one mechanical component. Proper operation of the mechanical component would be indicated by a consistent pattern of imaging, thermal, acoustic, and motion detections. A change in this consistent pattern would thereby equate to a potential issue with the operation of the mechanical component, such as indications of a failure, a leak, or the like.

(104) In the example embodiment shown in FIGS. **8A-8B** and **9A-9B**, four sensors **205** are illustrated as being positioned on each blender transport **110**. However, this may not necessarily be the case. The number, type, position, angle, and other characteristics of sensors **205** are not intended to be limiting to that shown in the drawings, and may be determined based on the use case (e.g., what equipment, component, or apparatus of the blender transport is being monitored), and the types of predetermined states (e.g., what are the relevant operational states of the monitored equipment) that are being monitored by sensors **205**.

(105) As one non-limiting example, sensors **205** may be imaging sensors that are positioned to face blender **806** in FIGS. **8A-8B** (or hoppers **904** in FIGS. **9A-9B**) to detect sand (or other additive) flow rates and to detect whether sand is overflowing out of blender **806** or hopper **904**. When an overflow or failure begins, the consistent nature of the imaging will change as the imaging sensor **205** detects sand visibly overflowing out of blender **806** or hopper **904**, fracing fluid overflowing out of blender **806**, smoke emanating from blender **806**, hopper **904** or from equipment surrounding these components, and the like.

(106) For example, as shown in FIGS. **8A-8B** and **9A-9B**, one or more sensors **205** (e.g., cameras) may be positioned on blender transport **110** to provide a video feed of blenders **806**, hoppers **904**, or blending tubs. This sensor data (e.g., image data, laser signal measurement data, flow sensor data, IR data, and the like) may be output to control center **605** that may be configured with an underlying AI model **630** that is created and deployed for this blender transport **110** use case, and that may be able to detect predetermined states corresponding to the equipment being monitored on blender transport **110**, and perform corresponding predetermined operations or functions based on the detected state. As explained previously, the predetermined operations corresponding to the detectable predetermined states of AI model **630** corresponding to the blender transport **110** use case may similarly be preconfigured (and be reconfigurable) by the operator. Control center **605** may thus configure control system **660** of blender transport **110** to perform predetermined operations (e.g., notify operator, change sand flow rate, shut down blender transport **110**) based on the predetermined state detected by AI model **630** based on sensor **205** data input to the model for the blender transport use case.

(107) In previous discussions, the AI system of the present disclosure has been described as monitoring and controlling mechanical components on the fracturing pump transport **108** and the blender transport **110**. As will be appreciated with the benefit of the present disclosure, other mechanical or electrical components of the hydraulic fracturing fleet **103** can also benefit from

monitoring and control provided by the AI system of the present disclosure. For example, AI system may be configured to compare a calculated horsepower output of a component or transport (e.g., frac pump transport **108**) of fleet **103** with a measured power consumed, and perform predetermined operations (e.g., notify operator) when the AI system detects inefficiency based on any deviation in the comparison being beyond a predetermined limit. As another example, the AI system may be configured to alert an operator (i.e., predetermined operation or function) when a user is detected (e.g., using one or more sensors **205**, and a corresponding trailed AI model **630**) to be within a predetermined radius of predetermined equipment (e.g., fluid end **216**, blender **665**, hopper **670**, and the like) that is determined to be in a predetermined state (e.g., overheating state, smoking state, fluid end packing failure (leaking) state, overflow state, and the like).

(108) FIG. **10** is a flowchart illustrating method **1000** for creating and implementing the AI model in accordance with one or more embodiments. Method **1000** may be performed for each AI model created and deployed for each use case (or corresponding apparatuses) on AI system **600**. As illustrated in FIG. **10**, method **1000** may include two phases. In phase-1 **1001**, control center **605** (or another computer system) may control one or more sensors **205** to capture sensor data offline or a priori having known responses (block **1010**). Alternately, sensor data captured at block **1010** may be historical data from past hydraulic fracturing operations that have been analyzed to identify corresponding known responses (e.g., analyze historical images to identify corresponding known predetermined states). For example, in the fluid end packing failure use case, the captured sensor data at block **1010** may be image data from one or more image sensors **205** placed at one or more fluid end assemblies **216** of one or more fracturing pump transports. Sensor data at block **1010** may include known data for each of a plurality of predetermined states in which the corresponding AI model being trained can classify future test data into.

(109) At block **1020**, control center **605** may use data captured at block **1010** to train an AI model. As explained in connection with FIG. **7**, AI model may be any artificial intelligence model (e.g., machine learning or deep learning model) that can detect predetermined states (e.g., perform prediction, classification, or diagnostic operation) for input test field data. At block **1030**, control center **605** may deploy trained AI model (e.g., AI model **630**) on AI system **600** for automatically monitoring and controlling equipment (e.g., components or apparatuses corresponding to the AI model) on one or more transports of fracturing system **103** for the use case for which AI model has been trained to detect the predetermined states.

(110) In phase-2 **1002**, one or more sensors **205** corresponding to the use case of the AI model deployed at block **1030** may capture sensor data continuously and in real-time during hydraulic fracturing operations (block **1040**). For example, sensor data may be image data that is captured by one or more image sensors **205** in real-time at a predetermined number of frames per second so as to capture a live video feed of a plurality of apparatuses (e.g., plurality of fluid end assemblies **216** on plurality of frac pump transports **108**). As another example, sensor data can be audio data (that can be analyzed by a corresponding AI model), thermal imaging data (that can be analyzed by a corresponding AI model), and the like.

(111) At block **1050**, sensor data captured at block **1040** may be input to the trained and deployed AI model **630** on control center **605** for the AI model to perform the state detection operation for each apparatus of the use case based on corresponding input sensor data. At block **1060**, the AI model may detect one of a plurality of predetermined states as a predetermined state corresponding to the sensor data input at block **1050** and output information regarding the detected predetermined state to control unit **640**. Thus, at block **1060**, the AI model may detect the current predetermined state of each of the plurality of apparatuses that are being monitored and controlled for the corresponding use case and output the detected state for each apparatus based on corresponding received sensor data. For example, the AI model may detect the state of Pump A on Trailer **1** based on Pump A sensor data and detect the state of Pump B on Trailer **2** based on Pump B sensor data. At block **1070**, control unit **640** may perform one or more predetermined operations corresponding

to the predetermined state detected at block **1060**. For example, control unit **640** may perform a first operation (e.g., notify operator) for Pump A on Trailer **1** based on the detected state thereof, and perform a second operation (e.g., automatically shut down pump) for Pump B on Trailer **2** based on the detected state thereof. As another example, control unit **640** may further perform a third operation (e.g., issue an alert) for Pump B on Trailer **2** based on detection of a user within a predetermined radius (“danger zone”) of Pump B while Pump B is in the detected state (e.g., overheating state, high-pressure fluid leak state, and the like). Steps performed at blocks **1040-1070** may be repeatedly performed based on sensor data captured continuously and in real-time at block **1040** for each of a plurality of components (or equipment) corresponding to the use case associated with the deployed AI model. In addition, as explained in connection with FIG. 7, control unit **640** may also update AI model **630** deployed at block **1030** based on new sensor data captured at block **1040** (block **1080**).

(112) FIG. 11 shows an illustrative simplified block diagram of computing system **1100** that includes one or more trained and deployed AI models **630** in accordance with one or more embodiments. Computing system **1100** may correspond to or may be part of a computer and/or any other computing device, such as a workstation, server, mainframe, super computer, and/or portable computing device. With reference to FIGS. 1, 2A-2B, 6, 7, 8A-8B, 9A-9B, computing system **1100** may correspond to a computer implemented in data van **114**, control system **208**, control center **605**, control system **603**, control system **660**, and/or control system **804**. Computing system **1100** comprises a processor **1102**, which may be also be referenced as a central processor unit (CPU). Processor **1102** (e.g., programmable control device) may communicate (e.g., via a system bus **1170**) and/or provide instructions to other components within the computing system **1100**, such as the input interface **1104**, output interface **1106**, and/or memory **1108**. In one embodiment, processor **1102** may comprise one or more multi-core processors and/or memory mediums (e.g., cache memory) that function as buffers and/or storage for data. Additionally, processor **1102** may be part of one or more other processing components, such as application specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and/or digital signal processors (DSPs). Although FIG. 11 illustrates that processor **1102** may be a single processor, processor **1102** is not so limited and instead may represent multiple processors. The processor **1102** may be configured to implement any of the operations described herein, including operations **700** and **1000** as described in FIGS. 7 and 10, and operations performed by AI system **600** described in FIG. 6.

(113) FIG. 11 illustrates that memory **1108** may be operatively coupled to processor **1102**. Memory **1108** may be a non-transitory medium or non-transitory program storage device configured to store various types of data. For example, memory **1108** may include one or more memory devices that comprise secondary storage, read-only memory (ROM), and/or random-access memory (RAM). The secondary storage is typically comprised of one or more disk drives, optical drives, solid-state drives (SSDs), and/or tape drives and is used for non-volatile storage of data. In certain instances, the secondary storage may be used to store overflow data if the allocated RAM is not large enough to hold all working data. The secondary storage may also be used to store programs that are loaded into the RAM when such programs are selected for execution. The ROM is used to store instructions and perhaps data that are read during program execution. The ROM is a non-volatile memory device that typically has a small memory capacity relative to the larger memory capacity of the secondary storage. The RAM is used to store volatile data and perhaps to store instructions.

(114) As shown in FIG. 11, memory **1108** may be used to house the instructions and models for carrying out various embodiments described herein. In an embodiment, memory **1108** may comprise one or more trained and deployed AI models **630** and instructions for implementing operations **700** and **1000** as described in FIGS. 7 and 10, and operations performed by AI system **600** described in FIG. 6. Instructions and data in memory **1108** may be accessed and implemented by processor **1102**. Additionally, or alternatively, AI models **630**, instructions for implementing operations **700** and **1000** as described in FIGS. 7 and 10, and operations performed by AI system

600 described in FIG. 6 may be stored and accessed within memory embedded in processor **1102** (e.g., cache memory). In one embodiment, memory **1108** may interface with system bus **1170** (e.g., a computer bus) to communicate and/or transmit information stored in memory **1108** to processor **1102** during execution of software programs, such as software applications that comprise program code, and/or computer executable process steps that incorporate functionality described herein.

(115) Persons of ordinary skill in the art are aware that software programs may be developed, encoded, and compiled in a variety of computing languages for a variety of software platforms and/or operating systems and subsequently loaded and executed by processor **1102**. In one embodiment, the compiling process of the software program, may transform program code written in a programming language to another computer language such that the processor **1102** is able to execute the programming code. For example, the compiling process of the software program may generate an executable program that provides encoded instructions (e.g., machine code instructions) for processor **1102** to accomplish specific, non-generic, particular computing functions, such as detecting predetermined states of a particular use case based on a corresponding trained AI model, and performing predetermined operations or functions based on the detected predetermined state.

(116) After the compiling process, one or more of AI models **630**, instructions for implementing operations **700** and **1000** as described in FIGS. 7 and 10, and operations performed by AI system **600** described in FIG. 6 may be loaded as computer executable instructions or process steps or data to processor **1102** from storage (e.g., memory **1108**, storage medium/media, removable media drive, and/or other storage device) and/or embedded within the processor **1102**. Processor **1102** can execute the stored instructions or process steps or data to perform instructions or process steps to transform computing system **1100** into a non-generic, particular, specially programmed machine or apparatus. Stored data, e.g., data stored by a storage device, sensor data, and the like can be accessed by processor **1102** during the execution of computer executable instructions or process steps to instruct one or more components within computing system **1100**.

(117) Alternatively, rather than programming and/or loading executable instructions onto memory **1108** and/or processor **1102** to form a non-generic, particular machine or apparatus, persons of ordinary skill in the art are aware that stored instructions may be converted and implemented as hardware customized for a particular use. In one embodiment, implementing instructions, such as detecting predetermined states of a particular use case based on a corresponding trained AI model and input sensor data, and performing predetermined operations based on the detected predetermined state, by loading executable software into a computing device, can be converted to a hardware implementation by well-known design rules. For example, the compiling process of the software program may build a sequence of instruction bits that control and arrange a sequence of control gate-level components that write data onto buses, into latches and registers, across channels, memory, and/or other components of processor **1102** and/or memory **1108**. The compiling of one or more of AI models **630**, instructions for implementing operations **700** and **1000** as described in FIGS. 7 and 10, and operations performed by AI system **600** described in FIG. 6 may produce gate-level components with fixed relationships designed to accomplish specific, non-generic, particular computing functions.

(118) The decisions between implementing a concept in software versus hardware may depend on a number of design choices that include stability of the design and numbers of units to be produced and issues involved in translating from the software domain to the hardware domain. Often a design may be developed and tested in a software form and subsequently transformed, by well-known design rules, to an equivalent hardware implementation in an ASIC or other application specific hardware that hardwires the instructions or process steps of the software. In the same manner as a machine controlled by a new ASIC is a non-generic, particular, specially programmed machine or apparatus, likewise a computing device (e.g., a computer) that has been programmed and/or loaded with executable instructions or process steps should be viewed as a non-generic,

particular, specially programmed machine or apparatus.

(119) FIG. 11 also illustrates that processor 1102 may be operatively coupled to input interface 1104 configured to receive input sensor data and/or direct-report data, and output interface 1106 configured to output and/or display an avatar model and/or perceptual data. The input interface 1104 may be configured to obtain input sensor data and/or direct-report data and/or other information via cables, connectors, wireless connections and/or other communication protocols. In one embodiment, the input interface 1104 may be a network interface that comprises multiple ports configured to receive and/or transmit data via a network. In particular, the network interface may transmit the data via wired links, wireless link, and/or logical links. Other examples of input interface 1104 may be universal serial bus (USB) interfaces, CD-ROMs, DVD-ROMs and/or connections to one or more sensors. Output interface 1106 may include to one or more connections for a graphic display (e.g., monitors), a printing device that produces hard-copies of the generated results, and/or a plurality of ports that transmit data via cables, connectors, wireless connections, and/or other communication protocols. FIG. 11 also illustrates that processor 1102 may be operatively coupled to one or more sensors 205 to generate the sensor data corresponding to fracturing system 103. Persons of ordinary skill in the art are aware that computing system 1100 may comprise other components well known in the art, such as other sensors, powers sources, and/or analog-to-digital converters, not explicitly shown in FIG. 11.

(120) Referring to FIG. 12, illustrative network architecture 1200 in which AI system 600 may be implemented includes a plurality of networks 1205, (e.g., 1205A, 1205B and 1205C), each of which may take any form including, but not limited to, a local area network (LAN) or a wide area network (WAN) such as the Internet. Further, networks 1205 may use any desired technology (wired, wireless or a combination thereof) and communication protocol (e.g., TCP, or transmission control protocol and PPP, or point to point). Coupled to networks 1205 are data server computer systems 1210 (e.g., 1210A and 1210B) that are capable of communicating over networks 1205. Also coupled to networks 1205, and/or data server computer systems 1210, are client or end-user computer systems 1215 (e.g., 1215A, 1215B and 1215C). Each of these elements or components may be a computer system or electronic device as described above with respect to FIGS. 1-11. In some embodiments, network architecture 1200 may also include network printers such as printer 1220 and network storage systems such as 1225. To facilitate communication between different network devices (e.g., server computer systems 1210, client computer systems 1215, network printer 1220 and storage system 1225), at least one gateway or router 1230 may be optionally coupled there between.

(121) At least one embodiment is disclosed and variations, combinations, and/or modifications of the embodiment(s) and/or features of the embodiment(s) made by a person having ordinary skill in the art are within the scope of the disclosure. Alternative embodiments that result from combining, integrating, and/or omitting features of the embodiment(s) are also within the scope of the disclosure. Where numerical ranges or limitations are expressly stated, such express ranges or limitations may be understood to include iterative ranges or limitations of like magnitude falling within the expressly stated ranges or limitations (e.g., from about 1 to about 10 includes, 2, 3, 4, etc.; greater than 0.10 includes 0.11, 0.12, 0.13, etc.). The use of the term “about” means $\pm 10\%$ of the subsequent number, unless otherwise stated.

(122) Use of the term “optionally” with respect to any element of a claim means that the element is required, or alternatively, the element is not required, both alternatives being within the scope of the claim. Use of broader terms such as comprises, includes, and having may be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of. Accordingly, the scope of protection is not limited by the description set out above but is defined by the claims that follow, that scope including all equivalents of the subject matter of the claims. Each and every claim is incorporated as further disclosure into the specification and the claims are embodiment(s) of the present disclosure.

(123) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(124) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise.

(125) Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention therefore should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.”

Claims

1. A method, comprising: accessing a thermal image of a component provided on a transport of a hydraulic fracturing fleet; inputting the thermal image into an artificial intelligence (AI) model, wherein the AI model is trained using a training set including a plurality of historical thermal images of historical components on transports of hydraulic fracturing fleets, each of the plurality of historical thermal images in the training set being labeled with one of a plurality of predetermined operating states for a corresponding historical component; detecting, with the AI model and based on the input thermal image, one of the plurality of predetermined operating states for the component; and operating the hydraulic fracturing fleet based on the detected predetermined operating state.
2. The method of claim 1, wherein the transport is a frac pump transport.
3. The method of claim 2, wherein the component is a fluid end of a frac pump mounted on the frac pump transport.
4. The method of claim 2, wherein the component is a transformer configured to perform a voltage step down operation.
5. The method of claim 2, wherein the component is an electric prime mover configured to drive a frac pump mounted on the frac pump transport.
6. The method of claim 1, wherein the transport is a blender transport.
7. The method of claim 6, wherein the component is a transformer configured to perform a voltage step down operation on the blender transport.
8. The method of claim 6, wherein the component is an electric prime mover configured to perform a blending operation on the blender transport.
9. The method of claim 1, wherein the plurality of predetermined operating states include an overheating state, a smoking state, and a leaking state.
10. The method of claim 1, wherein operating the hydraulic fracturing fleet comprises at least one of alerting an operator, or shutting down equipment.
11. A system, comprising: a thermal imaging sensor; memory storing an artificial intelligence (AI) model; and one or more processors operatively coupled to the thermal imaging sensor and the memory, the one or more processors being configured to: operate the thermal imaging sensor to capture a thermal image of a component provided on a transport of a hydraulic fracturing fleet; input the thermal image into the AI model, wherein the AI model is trained using a training set

including a plurality of historical thermal images of historical components on transports of hydraulic fracturing fleets, each of the plurality of historical thermal images in the training set being labeled with one of a plurality of predetermined operating states for a corresponding historical component; detect, with the AI model and based on the input thermal image, one of the plurality of predetermined operating states for the component; and operate the hydraulic fracturing fleet based on the detected predetermined operating state.

12. The system of claim 11, wherein the transport is a frac pump transport.

13. The system of claim 12, wherein the component is a fluid end of a frac pump mounted on the frac pump transport.

14. The system of claim 12, wherein the component is a transformer configured to perform a voltage step down operation.

15. The system of claim 12, wherein the component is an electric prime mover configured to drive a frac pump mounted on the frac pump transport.

16. The system of claim 11, wherein the transport is a blender transport, and wherein the component is a transformer configured to perform a voltage step down operation on the blender transport.

17. The system of claim 11, wherein the transport is a blender transport, and wherein the component is an electric prime mover configured to perform a blending operation on the blender transport.

18. The system of claim 11, wherein the plurality of predetermined operating states include an overheating state, a smoking state, and a leaking state.

19. The system of claim 11, wherein the one or more processors being configured to operate the hydraulic fracturing fleet comprises the one or more processors being configured to at least one of alert an operator, or shutdown equipment.

20. A frac pump transport, comprising: a frac pump comprising a fluid end packing; an image sensor; memory storing an artificial intelligence (AI) model; and one or more processors operatively coupled to the image sensor and the memory, the one or more processors being configured to: operate the image sensor to capture an image of the fluid end packing of the frac pump; input the image into the AI model, wherein the AI model is trained using a training set including a plurality of historical images of historical fluid end packings of frac pumps on frac pump transports of hydraulic fracturing fleets, each of the plurality of historical images in the training set being labeled with one of a plurality of predetermined operating states for a corresponding historical fluid end packing; detect, with the AI model and based on the input image, one of the plurality of predetermined operating states for the fluid end packing; and operate the frac pump transport based on the detected predetermined operating state.
